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Evidence in favour of density wave theory through age gradients observed in star formation history maps and spatially resolved stellar clusters

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ABSTRACT

Quasi-stationary density wave theory predicts the existence of an age gradient across the spiral arms with a phase crossing at the corotation radius. We have examined evidence for such age gradients using star formation history (SFH) maps derived from LIGHTNING, a spectral energy distribution fitting procedure, and by using spatially resolved stellar clusters. Three galaxies from the LEGUS survey were used to analyse the azimuthal offsets of spatially resolved stellar clusters. Kernel density estimation plots of azimuthal cluster distance offsets reveal prominent central peaks and secondary peaks on the positive side, relative to the density wave for NGC 5194 and NGC 5236. These secondary downstream peaks in the cluster distributions show overall evidence for an age gradient. NGC 628 shows secondary peaks on both sides of the density wave. The cluster distributions also show an increasing spatial spread with age, consistent with the expectation that they were born in the density wave. SFH maps of 12 nearby galaxies were analysed using SPIRALITY, a MATLAB-based code, which plots synthetic spiral arms over FITS images. The SFH maps reveal a gradual decrement (tightening) in pitch angles with increasing age. By analysing the pitch angle differences between adjacent age bins using the error function, the average of the probabilities shows a 69 per cent ± 25 per cent chance that the pitch angle values decrease (tighten) with increasing age. Thus, we see a tightening of the spiral pattern in galaxies, both when segregating stellar populations specifically by age or more generally by colour, as was shown in our previous studies.

Key words: galaxies: evolution – galaxies: fundamental parameters – galaxies: kinematics and dynamics – galaxies: spiral – galaxies: star clusters: general – galaxies: structure.

1 INTRODUCTION

The density wave theory (Lin & Shu 1964, 1966; Lin, Yuan & Shu 1969) explains the nature of spiral structures in galaxies as the result of long-lived, quasi-stationary density waves propagating through the galactic disc. These density waves induce shock fronts on the disc material, forcing them to form regions of higher gas density (Roberts 1969). When the densities of these regions exceed the Jean's criteria (Jeans 1902), star formation takes place, producing stars and stellar clusters. These newly formed stars and clusters drift away in an azimuthal direction from their initial birth places, producing an age gradient across the spiral arms, as a result of differential rotation in the disc. More recently, it has been proposed that density waves do not form semipermanent structures because of dissipation in the disc (Sellwood & Carlberg 1984). However, there remains the possibility that density waves are not truly transient (i.e. lasting only a rotation or two of the galactic disc) but may last for more than a half-dozen rotations (Sellwood 2012; Sellwood & Carlberg 2014).

Different stages of this process ought to be visible in different wavelengths of light. For instance, blue and ultraviolet light would show where newly born stars are found, just downstream from the star-forming region, while far-infrared light might show the early stages of the star formation. One long-standing claim is that the position of the density wave itself can be observed by viewing it in near-infrared light, which is sensitive to old/red disc stars which, if they are sufficiently well coupled to the density wave, can be expected to be crowded closer together where the density wave is found than in other regions of the disc. It is possible, depending on the value of the reduction factor (see equations 1 and 2 from Davis et al. 2015, which were originally presented in Lin & Shu 1966), for the density wave to be strong enough to cause collapse in gas clouds without strongly affecting the stellar disc itself.

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Nevertheless, it is widely reported that one can observe the density wave in this near-infrared light (Elmegreen & Elmegreen 1984). Note, however, that this means that there are two different gradients predicted by the theory. A kinematic colour gradient and an age gradient. The kinematic colour gradient is visible due to the contrast between the old, red disc stars' density, enhanced close to the density wave, and the newly born stars at the shockwave. These old stars have revolved around the disc a few times and they are typically a few billion years old.

For the purpose of this paper, we will only be focusing on the aging of the stellar populations (the age gradient). Stars of different masses, born together at the density wave shock front, drift away with time in the direction of the disc flow. Short-lived massive blue stars die away close to the edges of the spirals while low-mass red stars drift away downstream, creating a blue-to-red age gradient.

The key difference between the 'kinematic colour gradient' and the 'age gradient' is the extreme age difference between the two different red spiral arms. The age gradient should depict a range of stars, extending downstream of the density wave, lined up from blue to red in order of age since they were born. Therefore, an analysis of relatively young stars by position and by age may reveal the existence of this age gradient predicted by the theory.

The search for age gradients has yielded a variety of results (Schweizer 1976; Talbot, Jensen & Dufour 1979; Cepa & Beckman 1990; Hodge, Jaderlund & Meakes 1990), where some claimed to see the age gradients, while others did not. The existence of either the colour or the age gradient, or both, may ultimately shed light on the question of the transience of spiral-arm structure, since it is certainly expected that long-lived spiral arms should exhibit gradients of this type.

Martínez-García, González-López-lira & Bruzual-A (2009) claim to have seen these age gradients in identified regions belonging to ten galaxies in a sample of 13 spiral galaxies of types A and AB (galaxies with no or weak bars, respectively). They arrived at this conclusion by using observations of dust, gas compression, molecular clouds within the neighbourhood of the spiral structure, and by using a photometric index *Q(rJgi)* to trace star formation. They have consistently found such age/colour gradients in their work (Martínez-García & González-López-lira 2015) and have even measured spiral pattern speeds by comparing the observations with stellar population synthesis models. Their findings favour quasi-stationary, long-lived spiral structures as proposed by the original Lin & Shu (1964) study.

More recently, Peterken et al. (2019) have observationally confirmed quasi-stationary density waves via analysis of the pattern speed in UGC 3825. Their spectral mapping of stellar ages allowed them to measure the offsets between young stars and their birth sites, in order to directly measure the pattern speed as a function of galactic radii. Their novel method has shown that, for UGC 3825, the pattern speed varies little with radius, and is thus consistent with the presence of a quasi-stationary density wave.

Sánchez-Gil et al. (2011) have also found convincing evidence of age gradients for some galaxies in their sample using a pixel-based analysis of the ionized gas emission. Based on burst ages derived from the Hα-to-far-ultraviolet flux ratio, they have derived age maps showing a wide range of patterns. They have concluded that generally grand designs exhibit more clear age patterns while non-grand designs, based on their individual circumstances, fail to demonstrate the patterns. They further claim that the cases with an absence of age patterns are simply instances where spiral density wave theory is not the dominant driver in the star formation process.

Using stellar cluster catalogues of three LEGUS[1] (Legacy Extragalactic UV Survey) galaxies, Shabani et al. (2018) have found yet again, varying results, where one galaxy, NGC 1566 showed strong age gradients, while NGC 5194 did not show a promising age trend. NGC 628, due to its weak spiral structure, also exhibited no clear evidence of an age pattern. It is important to note that in their study, they have used azimuthal offsets of the clusters with respect to the dark dust lanes on *B*-band images.

Choi et al. (2015) also claim that age gradients are not visible in M81, and therefore argue that M81 is not driven by quasi-stationary density waves, but rather by kinematic spiral patterns that are likely influenced by tidal interactions with the companion galaxies. They have used star formation histories (SFHs) of 20 regions around the spiral structure. For each region, they have used resolved stellar populations, thus producing spatially resolved SFHs.

In this paper, we will focus not on colour as a marker for stellar age, but directly on the ages of young stars in spiral arms. In this way, we hope to clearly establish the existence of an age gradient extending downstream from the star-forming regions. We use two methods to identify the positions of stars of different ages. One makes use of SFH maps provided by LIGHTNING (Eufrasio et al. 2017). The other relies on studies of young stellar clusters in nearby galaxies.

In Section 2, we present our sample of galaxies with SFH maps and those with spatially resolved stellar clusters. Section 3 is where we account our pitch angle measurements and azimuthal offsets of the clusters with respect to the spiral arms. For the penultimate Section 4, we provide the results of our SFH map and stellar cluster analyses. Finally, we present our summary and conclusions in Section 5, with a discussion of the implications of our findings. Throughout this work, all uncertainties are quoted at $1\sigma \approx 68.3$ per cent confidence intervals.

2 THE DATA SAMPLE

2.1 LIGHTNING outputs: SFH maps in non-parametric age bins

Using photometric data from *GALEX*, *Swift*, SDSS, 2MASS, *Spitzer*, *WISE*, and *Herschel*, we derived SFH maps with the help of LIGHTNING (Eufrasio et al. 2017), a spectral energy distribution (SED) fitting procedure. Each pixel depicts the SFH of that spatial region and has the units $M_\odot\,yr^{-1}$. The SFHs are derived in non-parametric bins of ages 0–10 Myr, 10–100 Myr, 0.1–1 Gyr, 1–5 Gyr, and 5–13.6 Gyr.

In a total sample of 37 nearby galaxies, only 12 galaxies (see Table 1) were selected for this study. The selection criteria included considerations on the morphology and the inclination angle (i). Six galaxies were rejected based on their morphologies (i.e. early-type or irregular galaxies), seven more were rejected due to having diffuse or indistinguishable arm features. The indistinguishable arm features were mainly due to low arm–interarm contrasts and pixel constraints in the SFH maps. Although NGC 3938 shows a clear spiral structure, due to its multiple arms and due to resolution constraints in the SFH maps, we were unable to clearly distinguish spiral features and hence, we excluded it from the selection. Again, 11 more galaxies were excluded due to high inclinations.

The inclination angle plays a pivotal role in the deprojection process; hence, it is crucial to select galaxies with moderate inclination angles ($i \lesssim 50^\circ$). Because galaxies are randomly orientated

[1]Calzetti et al. (2015), https://legus.stsci.edu/

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Table 1. Galaxy sample.

Galaxy name (1)	Morphology (2)	RA (J2000) (3)	Dec (J2000) (4)	Distance (Mpc) (5)	Position angle (°) (6)	Inclination (°) (7)
NGC 0628	SA(s)c	01 36 41.7	+15 47 01	6.7	25	7.0
NGC 1097	SB(s)b	02 46 19.05	−30 16 29.6	16.7	130	26.8
NGC 3031	SA(s)ab	09 55 33.2	+69 03 55	3.6	150	34.5
NGC 3184	SAB(rs)cd	10 18 16.86	+41 25 26.59	13.6	135	5.3
NGC 4254	SA(s)c	12 18 49.6	+14 24 59	32.3	24	9.2
NGC 4321	SAB(s)bc	12 22 54.8	+15 49 19	16.1	28	9.7
NGC 4725	SAB(r)ab	12 50 26.58	+25 30 02.90	21.6	35	20.1
NGC 5194	SA(s)bc	13 29 52.7	+47 11 43	8.9	172	21.9
NGC 5236	SAB(s)c	13 37 00.9	−29 51 56	4.5	45	14.8
NGC 5457	SAB(rs)cd	14 03 12.5	+54 20 56	7.4	29	3.9
NGC 6946	SAB(rs)cd	20 34 52.3	+60 09 14	3.8	53	20.0
NGC 7552	(R')SB(s)ab	23 16 10.7	−42 35 05	14.8	1	6.7

Notes. Columns: (1) Galaxy name. (2) Hubble morphological type. (3) RA (J2000), presented as h, min, and s. (4) Dec (J2000), presented as h, min, and s. (5) Distance (Mpc). (6) Position angle (°). (7) Inclination (°). All RA, Dec, and distance measurements presented here are based on accepted values found in the NASA/IPAC Extragalactic Database (NED) data base. Position angles (east of north) and inclinations are based on our IRAF ELLIPSE fitting measurements and were verified by checking against commonly accepted measurements found in the NED data base.

Table 2. Cluster samples for NGC 628, NGC 5194, and NGC 5236.

Cluster ID (1)	RA (deg) (2)	Dec (deg) (3)	log ($T_{\rm age}$; yr) (4)	$\Delta\Theta$ (rad) (5)
NGC 628-C001	24.169 12	15.803 96	$7.95^{+0.35}_{-1.11}$	0.331 23
NGC 628-C002	24.175 94	15.802 97	$7.04^{+0.13}_{-0.00}$	0.010 79
NGC 628-C003	24.176 87	15.802 70	$8.30^{+0.00}_{-0.30}$	0.027 75
NGC 5194-C001	202.400 41	47.130 25	8.06[a]	− 0.725 00
NGC 5194-C002	202.405 43	47.130 38	6.78[a]	− 1.039 14
NGC 5194-C003	202.457 79	47.132 46	6.00[a]	0.102 08
NGC 5236-C001	204.265 05	− 29.883 05	$9.10^{+0.00}_{-0.90}$	− 0.062 09
NGC 5236-C002	204.272 20	− 29.882 26	$9.30^{+0.40}_{-1.20}$	− 0.067 31
NGC 5236-C003	204.280 27	− 29.878 85	$9.00^{+0.10}_{-0.04}$	− 0.019 84

Notes. Columns: (1) Cluster ID. (2) RA (deg). (3) Dec (deg). (4) log ($T_{\rm age}$; yr). (5) Azimuthal offset of clusters relative to the spiral arm (rad). The data for this table were obtained from table 1 of Chandar et al. (2016), table 3 of Ryon et al. (2015, 2017).
[a]For these galaxies, no error values were mentioned in the original study (Chandar et al. 2016).
Only a portion of the table is shown here to demonstrate its content.

with respect to our line of sight, as the initial step, all the images had to be deprojected to a face-on orientation. Considering the commonly accepted inclination angles and position angles found in the literature (NED[2] data base) and using the traditional approach of fitting elliptical isophotes, using the IRAF[3] (Tody 1986, 1993) ELLIPSE routine (Jedrzejewski 1987), and transforming them to a circular configuration, we deproject each image to a face-on orientation.

2.2 Spatially resolved stellar clusters

We use position coordinates of spatially resolved stellar clusters for NGC 5194, as summarized in Chandar et al. (2016), and position coordinates of stellar clusters in NGC 628 and NGC 236, as summarized by Ryon et al. (2015, 2017). All the sources have age estimates, which we use to categorize the clusters into different age bins. Table 2 columns 2 and 3 depict the RA and Dec values of the clusters, while column 4 gives the estimated cluster age in a log scale. Fig. 1 shows the spatial distribution of the 3816 compact star clusters found in NGC 5194 from Chandar et al. (2016) in the age bins of 0–10 Myr, 10–100 Myr, 0.1–1 Gyr, 1–5 Gyr, and 5–13.6 Gyr. It also shows a closeup view of the northern arm of NGC 5194 with stellar clusters belonging to the same age bins. Ryon et al. (2015)'s sample for NGC 5236 contains a total of 478 clusters and Ryon et al. (2017)'s sample for NGC 628 contains a total of 320 clusters. Only a selected set of clusters were used from the original samples based on the age constraints.

3 ANALYSIS

3.1 Measuring pitch angles of the SFH maps (LIGHTNING outputs)

Once we have all our SFH maps deprojected to face-on orientations, we start measuring pitch angles. The pitch angle (ϕ) of a given spiral

[2]NED; Helou et al.(1991), https://ned.ipac.caltech.edu/
[3]IRAF (Tody 1986, 1993), https://iraf.net/

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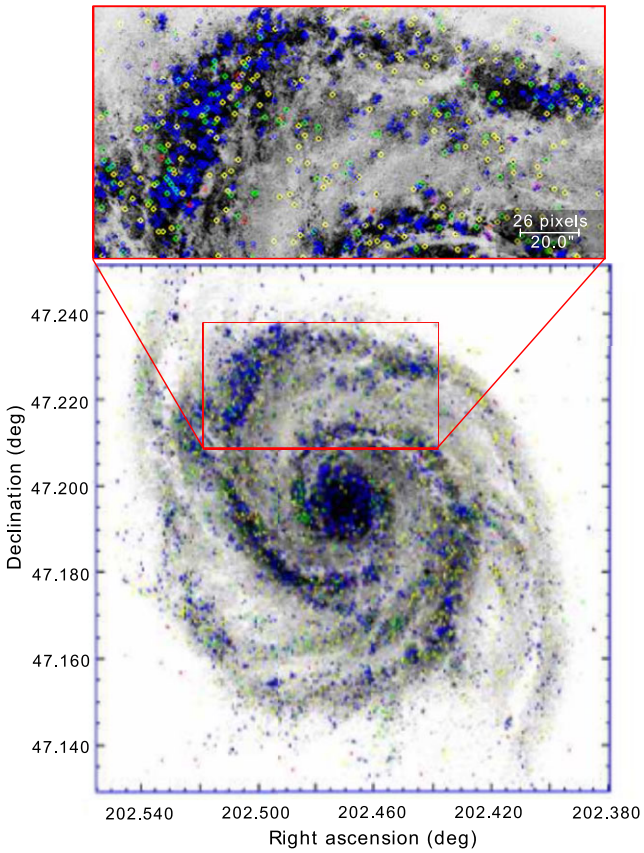

Figure 1. NGC 5194 with 3816 compact stellar clusters overlapped on an 8 μm image, along with a closeup view of the northern arm (above). Blue: 0–10 Myr. Green: 10–100 Myr. Yellow: 0.1–1 Gyr. Red: 1–5 Gyr. Magenta: 5–13.6 Gyr.

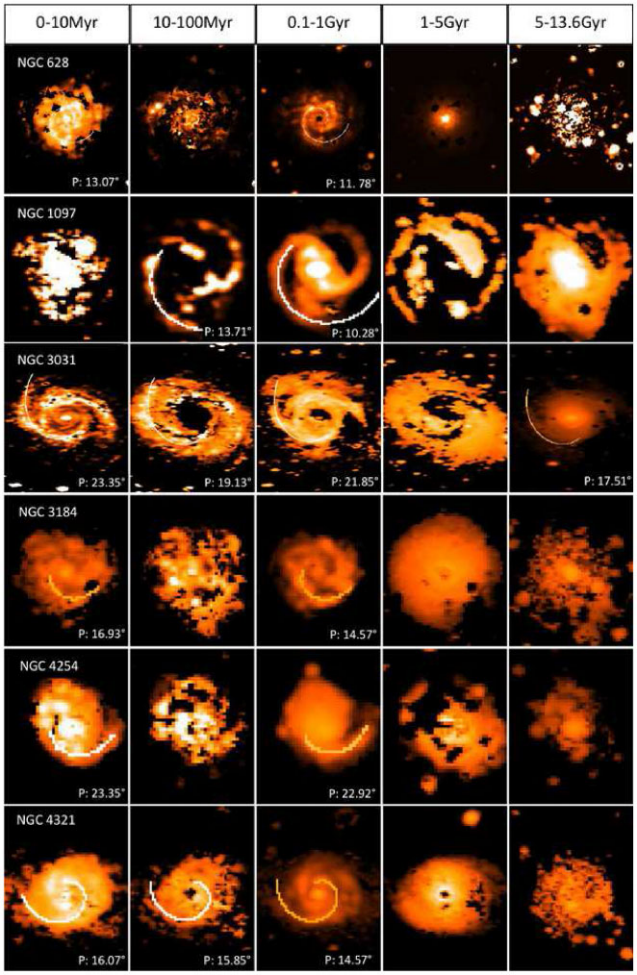

Figure 2. SPIRALITY: Arm trace outputs of SFH maps for six galaxies in five different age bins. Each pixel denotes the star formation activity in the region and has units of $M_{\odot}\ yr^{-1}$. The traces were included only on the maps that had measurable pitch angles.

is defined as the angle between the spiral's tangent at a given point and the tangent line drawn to a concentric circle that passes through that point.[4] Pitch angle measurements were performed by initially visually verifying using a PYTHON-based code (SPIRAL_OVERLAY).[5] The code generates a graphical interface that enables the user to load a FITS image along with a synthetic logarithmic spiral overlapped on a foreground layer. We can change the pitch angle, phase angle, and the number of arms of the synthetic spiral to match the features of the underlying actual image file. Once we have a rough estimate of the pitch angles, we use SPIRALITY[6] (Shields et al. 2015a,b), a MATLAB-based code, to measure pitch angles accurately. SPIRALITY uses a template-fitting approach by using a varying inner radius and focuses on a global best-fitting pitch angle.

The SPIRALITY, SPIRAL-ARM-COUNT (Shields et al. 2015a,b) script fits synthetic logarithmic spiral arms over the actual image spiral arms, based on image pixel brightness. The entire process is automated, hence minimizing the user bias in tracing these spirals. See Fig. 2 for the traced synthetic spirals. Only a portion of the spirals are displayed in the paper.

For those galaxies with clear image data, we used another independent method, namely 2DFFT (Puerari & Dottori 1992; Saraiva Schroeder et al. 1994; Puerari et al. 2000; Seigar et al. 2008; Davis et al. 2012, 2016), which uses a modified two-dimensional fast Fourier-transform process to measure the pitch angles for different harmonic modes. Upon verifying using these three methods, the final results were generated and are tabulated in Table 3. Table 3 shows pitch angle measurements for each SFH map belonging to five different age bins: 0–10 Myr, 10–100 Myr, 0.1–1 Gyr, 1–5 Gyr, and 5–13.6 Gyr. Obviously, pitch angle measurements are only reported for those age bin images in which spiral structure was visible. Since 8 μm images are tracers for gas and dust lanes, the brightest regions in the 8 μm images denote the approximate location of the density wave. Hence, we have also measured pitch angles of the 8 μm images for each of these galaxies. For some galaxies, the pitch angle measurements were adopted from our previous study (Abdeen et al. 2020).

[4]The sign of ϕ is merely representative of the direction of winding (i.e. clockwise or counterclockwise). In this work, we will always present the absolute value of pitch angle, $|\phi|$. For introductory reading on the pitch angle of logarithmic spirals, see section 2 from Davis, Graham & Seigar (2017).
[5]PYTHON OL script: https://github.com/ebmonson/2DFFTUtils-Module.
[6]SPIRALITY:(Shields et al. 2015a,b), http://ascl.net/phpBB3/download/file.php?id = 29.

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Table 3. Average pitch angle measurements of SFH maps.

Galaxy name (1)	ϕ° (0–10 Myr) (2)	ϕ° (10–100 Myr) (3)	ϕ° (0.1–1 Gyr) (4)	ϕ° (1–5 Gyr) (5)	ϕ° (5–13.6 Gyr) (6)	ϕ° (8.0 μm) (7)
NGC 0628	13.07 ± 0.23	–	11.78 ± 0.54	–	–	12.83 ± 0.66
NGC 1097	–	13.71 ± 0.68	10.28 ± 0.23	–	–	12.71 ± 0.32
NGC 3031	23.35 ± 0.68	19.13 ± 0.84	21.85 ± 0.31	–	17.51 ± 1.21	21.21 ± 2.31
NGC 3184	16.93 ± 2.13	–	14.57 ± 1.43	–	–	19.81 ± 2.30
NGC 4254	23.35 ± 2.21	–	22.92 ± 1.32	–	–	20.50 ± 1.73
NGC 4321	16.07 ± 2.78	15.85 ± 4.51	14.57 ± 1.46	–	–	14.73 ± 1.68
NGC 4725	12.86 ± 2.43	11.14 ± 3.24	–	–	–	11.92 ± 2.30
NGC 5194	12.00 ± 1.23	10.96 ± 1.45	11.36 ± 0.45	–	9.86 ± 2.13	10.48 ± 2.73
NGC 5236	13.28 ± 1.23	–	12.21 ± 0.56	–	–	16.74 ± 1.24
NGC 5457	28.92 ± 2.45	27.64 ± 4.56	25.07 ± 2.46	–	–	23.85 ± 2.71
NGC 6946	18.21 ± 3.98	–	16.93 ± 2.65	–	–	24.63 ± 2.81
NGC 7552	–	–	27.64 ± 5.84	–	–	28.99 ± 4.75

Notes. Columns: (1) Galaxy name. (2) Pitch angle in degrees for 0–10 Myr. (3) Pitch angle in degrees for 10–100 Myr. (4) Pitch angle in degrees for 0.1–1 Gyr. (5) Pitch angle in degrees for 1–5 Gyr. (6) Pitch angle in degrees for 5–13.6 Gyr. (7) Pitch angle in degrees for *Spitzer* 8.0 μm.

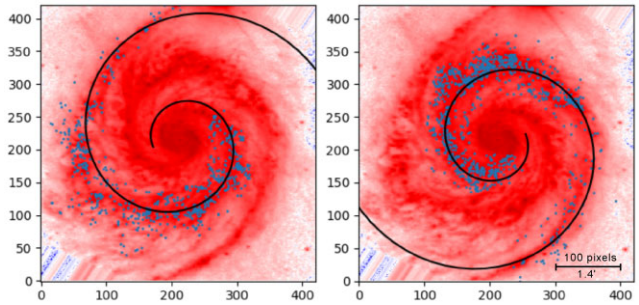

Figure 3. Left-hand panel: Stellar clusters (blue circles) in the vicinity of the northern spiral arm of NGC 5194. Right-hand panel: Stellar clusters in the vicinity of the southern spiral arm of NGC 5194. Synthetic logarithmic spirals of $\phi = 10^\circ\!.5$ are traced over each underlying 8 μm image.

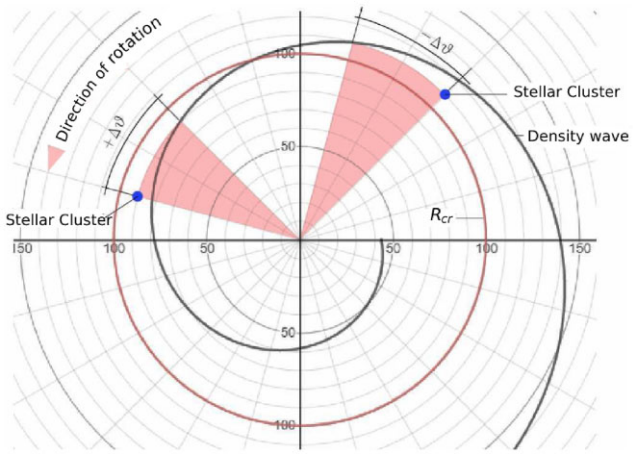

Figure 4. Schematic diagram representing the azimuthal offsets of clusters relative to the density wave.

3.2 Measuring the azimuthal offset of clusters, relative to the spiral arms

For the three galaxies for which cluster ages and positions are available, NGC 628, NGC 5194, and NGC 5236, we wished to compare the cluster positions with the rough position of the density wave. Accordingly, we located the minimum surface brightness locations of the arm–interarm regions in the 8 μm images in order to define the spiral-arm regions in these galaxies. Fig. 3 shows the northern and the southern arms of NGC 5194, along with the clusters belonging to each region. Fig. 4 is a schematic diagram representing azimuthal offsets of clusters relative to the density wave (synthetic logarithmic spiral traced over the 8 μm image spiral arm). Azimuthal offsets were measured for each cluster relative to the density wave (see column 5 in Table 2).

4 RESULTS

4.1 Results on the analysis of the SFH maps

While observing the pitch angle measurements in Table 3 and Fig. 5, it is apparent that, in general, the pitch angles decrease with increasing age. It is true that, for most of the cases, there are significant error bars, however, considering the overall sample in general, galaxies show a tendency to decrease their pitch angle with increasing age and it is visible in each case. This simple trend can be found in most of the SFH maps with few exceptions. In order to quantify this claim, we computed the pitch angle difference ($\phi_1^\circ - \phi_2^\circ$) and by using the error function we determined $P(\phi_1^\circ > \phi_2^\circ)$, the probability that $\phi_1^\circ > \phi_2^\circ$ (see Table 4). By observing the last column of Table 4, it is apparent that predominantly $\phi_1^\circ > \phi_2^\circ$. Considering the average of the entire sample for adjacent age bins, ϕ_2° is less than ϕ_1°, 69 per cent ± 25 per cent of the time. It is true that with the large error bars, this trend is not as clear as we would desire, but following this statistical investigation, we may undeniably conclude that predominantly $\phi_1^\circ > \phi_2^\circ$, hence the pitch angles decrease with increasing age.

It is also apparent that the 0.1–1 Gyr age bin has the clearest spiral structure, while none of the maps show clear spiral structures in the age bin of 1–5 Gyr (see Fig. 2 and Table 3). Only two galaxies, NGC 3031 and NGC 5194, had observable spiral features in the age bin of 5–13.6 Gyr. It is important to state that, an unobservable spiral structure in a particular age bin does not necessarily imply that the entire spiral structure of the galaxy disappears. It simply implies that the light produced by that fraction of stars does not form a visible spiral structure, yet there are other stars with different ages, continuously producing light.

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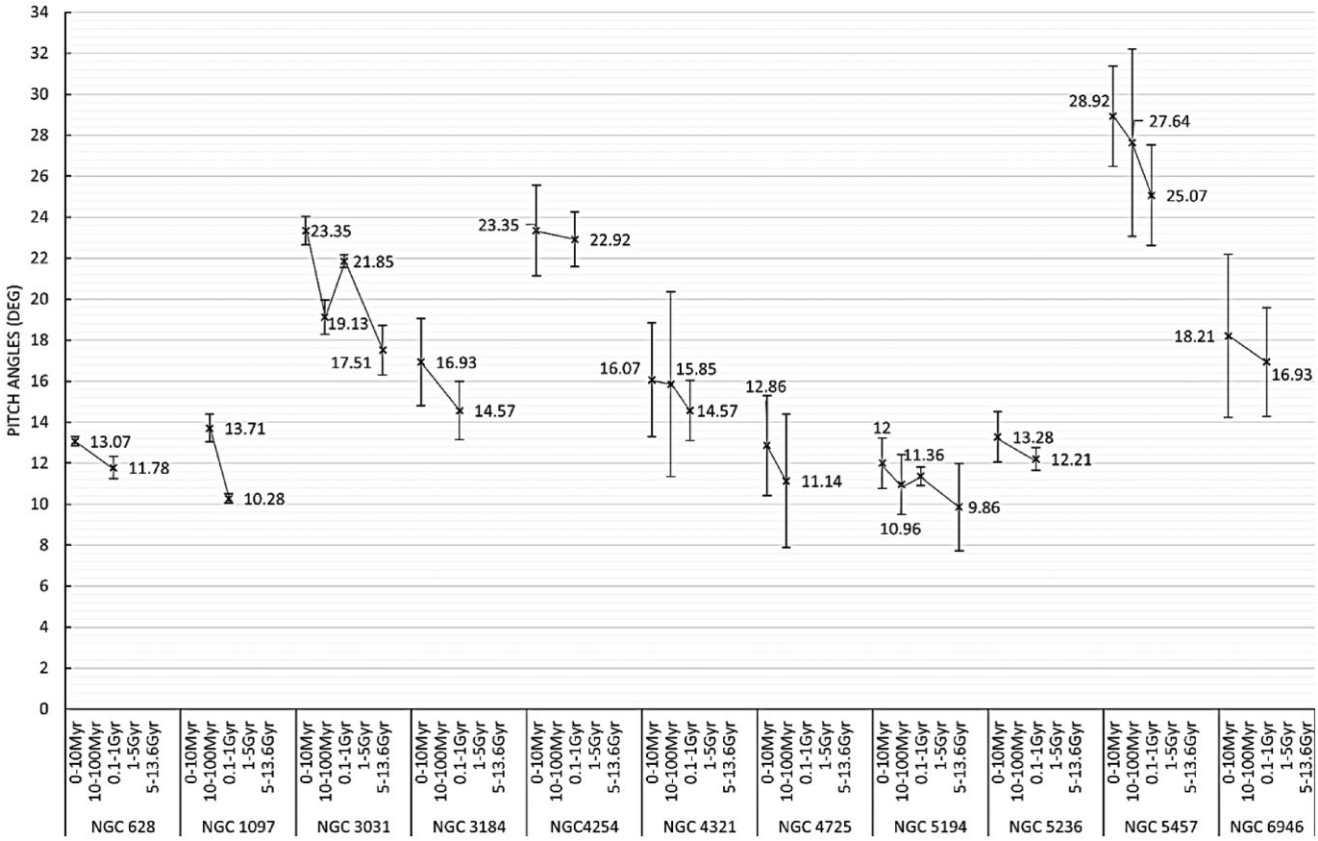

Figure 5. The gradual decrement of the pitch angles with increasing age is clearly visible for each galaxy. The pitch angle measurements were made on SFH maps.

Table 4. Comparison of pitch angle differences.

Galaxy name (1)	Age groups (2)	ϕ_1° (3)	E_1° (4)	ϕ_2° (5)	E_2° (6)	$\phi_1^\circ - \phi_2^\circ$ (7)	Error (8)	Z (9)	$P(\phi_1^\circ > \phi_2^\circ$; per cent) (10)
NGC 0628	0–10 Myr versus 0.1–1 Gyr	13.07	0.23	11.78	0.54	1.29	0.59	2.20	98.6
NGC 1097	10–100 Myr versus 0.1–1 Gyr	13.71	0.68	10.28	0.23	3.43	0.72	4.78	100.0
NGC 3031	0–10 Myr versus 10–100 Myr	23.35	0.68	19.13	0.84	4.22	1.08	3.90	100.0
	10–100 Myr versus 0.1–1 Gyr	19.13	0.84	21.85	0.31	−2.72	0.90	−3.04	0.1
	0.1–1 Gyr versus 5–13.6 Gyr	21.85	0.31	17.51	1.21	4.34	1.25	3.47	100.0
NGC 3184	0–10 Myr versus 0.1–1 Gyr	16.93	2.13	14.57	1.43	2.36	2.56	0.92	82.1
NGC 4254	0–10 Myr versus 0.1–1 Gyr	23.35	2.21	22.92	1.32	0.43	2.57	0.17	56.6
NGC 4321	0–10 Myr versus 10–100 Myr	16.07	2.78	15.85	4.51	0.22	5.30	0.04	51.7
	10–100 Myr versus 0.1–1 Gyr	15.85	4.51	14.57	1.46	1.28	4.74	0.27	60.6
NGC 4725	0–10 Myr versus 10–100 Myr	12.86	2.43	11.14	3.24	1.72	4.05	0.42	66.4
NGC 5194	0–10 Myr versus 10–100 Myr	12.00	1.23	10.96	1.45	1.04	1.90	0.55	70.8
	10–100 Myr versus 0.1–1 Gyr	10.96	1.45	11.36	0.45	−0.40	1.52	−0.26	39.6
	0.1–1 Gyr versus 5–13.6 Gyr	11.36	0.45	9.86	2.13	1.50	2.18	0.69	75.5
NGC 5236	0–10 Myr versus 0.1–1 Gyr	13.28	1.23	12.21	0.56	1.07	1.35	0.79	78.6
NGC 5457	0–10 Myr versus 10–100 Myr	28.92	2.45	27.64	4.56	1.28	5.18	0.25	59.8
	10–100 Myr versus 0.1–1 Gyr	27.64	4.56	25.07	2.46	2.57	5.18	0.50	69.0
NGC 6946	0–10 Myr versus 0.1–1 Gyr	18.21	3.98	16.93	2.65	1.28	4.78	0.27	60.6

Notes. Columns: (1) Galaxy name. (2) The two adjacent age groups we are comparing. (3) ϕ_1° = Pitch angle for 1st age bin. (4) E_1° = Error for ϕ_1°. (5) ϕ_2° = Pitch angle for 2nd age bin. (6) E_2° = Error for ϕ_2°. (7) Pitch angle difference $= \phi_1^\circ - \phi_2^\circ$. (8) Error $= \sqrt{E_1^2 + E_2^2}$. (9) $Z = (\phi_1^\circ - \phi_2^\circ)/\sqrt{E_1^2 + E_2^2}$. (10) Probability of $\phi_1^\circ > \phi_2^\circ$ equals $0.5[1 + \mathrm{erf}\,(Z/\sqrt{2})]$.

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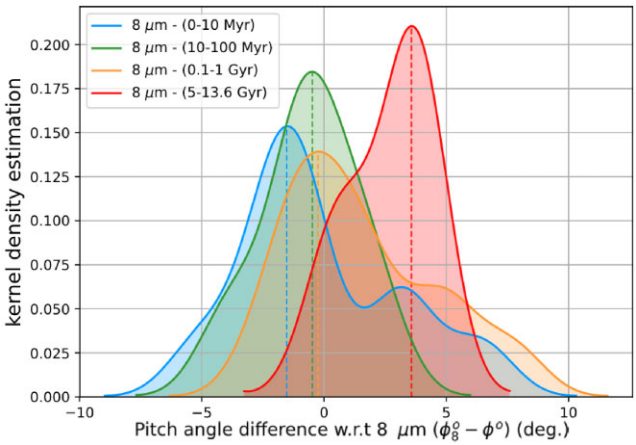

Figure 6. A KDE depicting the pitch angle differences relative to the 8 μm pitch angle measurements ($\phi_8^\circ - \phi^\circ$). Here, ϕ_8° denotes the pitch angle as measured in 8 μm and ϕ° is the pitch angle as measured in each age bin. The KDE of the pitch angle differences are analysed in four different age bins using a Gaussian kernel and an optimized bandwidth of 1°.296. Pitch angle differences tend to shift to a positive direction with increasing age. Dashed vertical lines coincide with the position of each peak.

Using the pitch angle measurements done for each age bin and the pitch angle measurements for the 8 μm images, we calculated the relative difference in pitch angles with respect to 8 μm ($\phi_8^\circ - \phi^\circ$). Here, ϕ_8° denotes the pitch angle as measured in 8 μm and ϕ° is the pitch angle as measured in each age bin. Since the 8 μm image is primarily an indication of dust and gas, it can denote the approximate location of the density wave; hence, the relative offset depicts how the pitch angle changes relative to the density wave for each age bin. It is important to note that a visible age gradient is not necessarily causally related to a visible pitch angle difference, but the two conditions have been observed to be highly coincident. From our previous studies (Abdeen et al. 2020), we have always seen pitch angle differences coinciding with visible arm centroid gradients. A kernel density estimation (KDE) is plotted to see the overall shift in the peak locations of the relative pitch angle distribution (see Fig. 6).

We further investigated this shift by calculating the relative pitch angle difference and by computing the probability that $\phi_8^\circ > \phi^\circ$. Here, ϕ_8° refers to the pitch angle of the 8 μm measurement while ϕ° refers to the other pitch angle measurements. In such, we computed the proportion of the normal distribution of their difference which is greater than zero. The probabilities were analysed in the different age bins and the results are as follows: $P(\phi_8^\circ > \phi_{0-10\ \mathrm{Myr}}^\circ) = 44$ per cent ± 32 per cent, $P(\phi_8^\circ > \phi_{10-100\ \mathrm{Myr}}^\circ) = 43$ per cent ± 25 per cent, $P(\phi_8^\circ > \phi_{0.1-1\ \mathrm{Gyr}}^\circ) = 63$ per cent ± 32 per cent, and $P(\phi_8^\circ > \phi_{5-13.6\ \mathrm{Gyr}}^\circ) = 75$ per cent ± 25 per cent. Accordingly, the probabilities that are <50 per cent indicate a predominance of negative pitch angle differences, e.g. $\phi_8^\circ - \phi_{10-100\ \mathrm{Myr}}^\circ < 0$ a majority of the time. While, the probabilities that are >50 per cent indicate a predominance of positive pitch angle differences, e.g. $\phi_8^\circ - \phi_{5-13.6\ \mathrm{Gyr}}^\circ > 0$ a majority of the time. This is precisely what we can see in Fig. 6 visually.

Using SFH maps of NGC 5194 and considering the brightest pixels (pixel values >0.0018 $M_\odot\ yr^{-1}$) of two age bins, 0–10 Myr and 10–100 Myr, we were able to overlay them on an 8 μm image to observe age trends. Fig. 7 represents the selected bright pixels in each age bin, the overlaid image, and a schematic diagram representing the age gradient. The red circle represents the location of the corotation radius (R_{cr}) as we measured in our previous study (Abdeen et al. 2020). It is important to note that inside the corotation radius, disc material is supposed to travel faster than the spiral arm, which is clearly visible in the overlaid image. Fig. 7 clearly shows the observational signature of an age gradient driven by a spiral density wave.

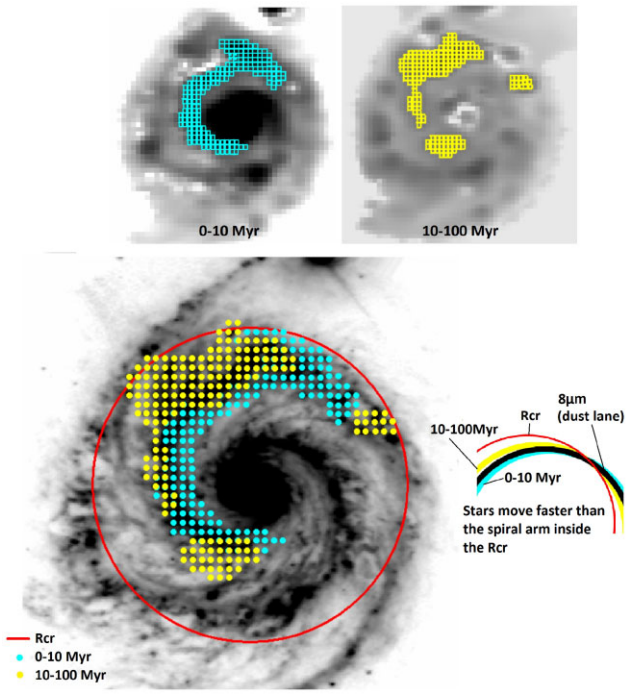

Figure 7. Top panels: SFH maps of NGC 5194 in the age the bins 0–10 Myr and 10–100 Myr. Bright pixels (pixel values >0.0018 $M_\odot\ yr^{-1}$) in the northern arm of NGC 5194 are coloured cyan (0–10 Myr age bin) and yellow (10–100 Myr age bin). Bottom panel: Locations of the individual bright pixels in each non-parametric bin are superimposed on a *Spitzer* 8 μm inverted colour image. It is important to note that stars move faster than the spiral-arm density wave inside the corotation radius; hence, an age gradient downstream is visible in certain stretches along the spiral arm.

4.2 Results on the analysis of the stellar clusters

Considering the azimuthal offsets of clusters relative to the density wave (synthetic logarithmic spiral location on the 8 μm images), we plotted histograms for each age bin corresponding to each spiral arm. Fig. 8 depicts the histograms plotted for NGC 5194, considering both of its spiral arms. It is important to note that there is a primary peak close to the density wave (0 rad in the histogram) and a secondary peak. It is possible that the primary peaks may be showing stars in the underlying density wave, which is closely coupled with the 8 μm dust lanes. Although we clearly see secondary peaks in each age bin, further investigation is necessary to clearly understand the exact reason for these peaks.

Fig. 9 shows a box plot (Tukey 1977) with the median locations of the overall cluster distributions for each galaxy. The box length denotes the interquartile range (IQR $\equiv Q_3 - Q_1$, where Q_1 and Q_3 are the first and third quartiles, respectively); the maximum and the minimum marks of the whiskers denote $Q_3 + 1.5 \cdot$ IQR and $Q_1 - 1.5 \cdot$ IQR, respectively. A clear shift in the mean locations is visible in NGC 5194, with the exception of the age bin 80–100 Myr.

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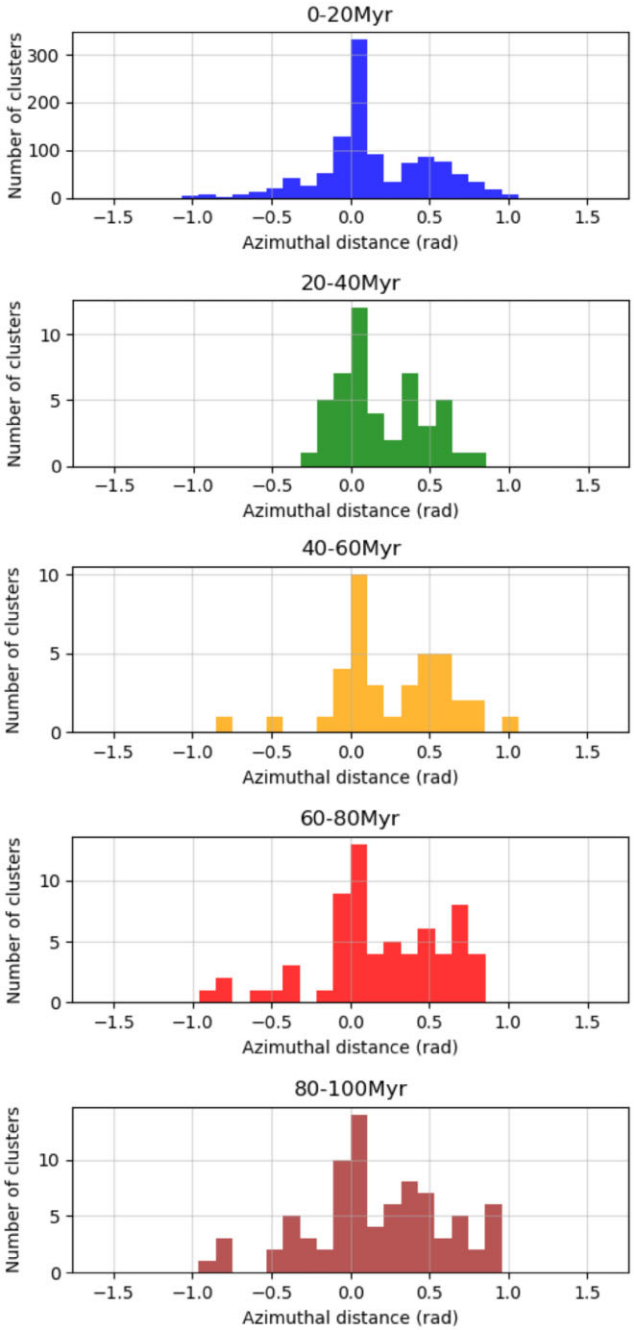

Figure 8. Histograms of the azimuthal cluster offsets of NGC 5194. They are categorized into five different age bins ranging from 0–100 Myr. It is important to note the location of the primary peak and the location of the secondary peak.

To better observe the age trends, KDEs were plotted using the python method SEABORN.KDEPLOT (Waskom 2021) employing a Gaussian kernel. For each graph, an average bandwidth (across all age bins) was selected using the Sheather & Jones method (Sheather & Jones 1991) performed individually on each age bin for that galaxy. Fig. 10 (top panel), shows NGC 628 in the age bins: 0–20 Myr, 20–40 Myr, 40–60 Myr, 60–80 Myr, and 80–100 Myr, plotted using a constant bandwidth of 0.112 rad. We analysed NGC 5194 (see Fig. 10, middle panel) in the same age bins with a bandwidth of 0.131 rad. For the case of NGC 5236, due to the lack of young

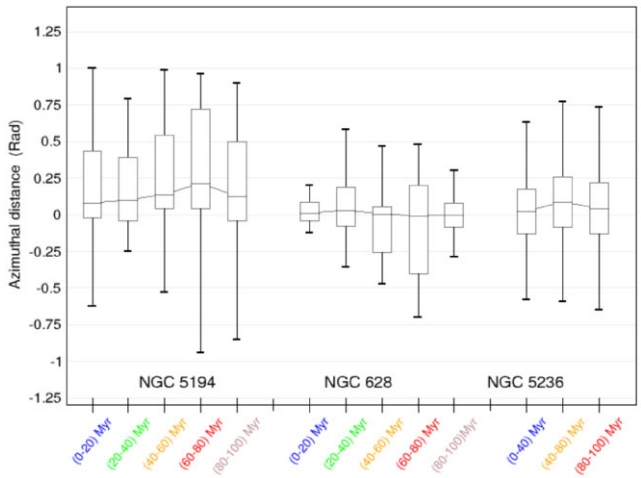

Figure 9. A box plot depicting the mean locations of the overall cluster distributions for each galaxy. The box length denotes the IQR, the maximum and the minimum marks of the whiskers denote $Q_3 + 1.5 \cdot \mathrm{IQR}$ and $Q_1 - 1.5 \cdot \mathrm{IQR}$, respectively.

clusters, we had to analyse the galaxy in age bins 0–40 Myr, 40–80 Myr, and 80–100 Myr, using a constant bandwidth of 0.113 rad.

The choices for the age bin sizes were made based on the availability of cluster data and also considering the age bins used for the SFH analysis.

Table 5 shows the locations of the peaks along with the full width at half-maximum (FWHM) of the primary peak. It is important to note that the FWHM increases gradually with the exception of NGC 628's 80–100 Myr age bin and NGC 5236's 80–100 Myr age bin. Most of the young clusters are concentrated close to the spiral arms, hence visible as a peak in the KDE graphs. As the clusters age, they migrate away from the spiral arm, thus increasing the FWHM with the cluster age. The gradual overall shift of the secondary peak provides us the direct evidence of an age gradient. To further visualize this age trend, vertical dashed lines were added to the peaks of the KDE.

Kolmogorov–Smirnov (KS) tests (Kolmogoroff 1933; Smirnov 1944) were performed to verify and to quantify whether clusters belonging to each age bin came from the same parent population. Assuming our null hypothesis (H_0): The data considered are from the same continuous distribution, for NGC 5194, the comparison between the age bins 0–20 Myr versus 60–80 Myr, resulted in the lowest D and p-value of 0.15978 and 0.08339, respectively. We reject the null hypothesis at the 8.34 per cent significance level. For NGC 628, the comparison between the age bins 0–20 Myr versus 20–40 Myr, resulted in the lowest D and p-value of 0.20735 and 0.349, respectively, where we reject the null hypothesis at 34.9 per cent. In the case of NGC 5236, the lowest values were $D = 0.20113$ and p-value $=0.2032$ for the age bin comparisons 0–40 Myr and 40–80 Myr, where we reject the null hypothesis at 20.32 per cent. Therefore, none of the KS tests provide convincing evidence (i.e. p-value <0.05) that older clusters come from different parent populations than younger clusters.

The shift in the secondary peak is clearly visible in all three of the galaxies, with few exceptions in certain age bins. It is important to note that both NGC 5194 and NGC 5236 show similar prominent secondary peaks on the positive side, while NGC 628 shows prominent secondary peaks on both sides of the spiral pattern. The exact reason for this double-peak distribution in NGC 628 should be further investigated. It is possible that the cluster samples may be

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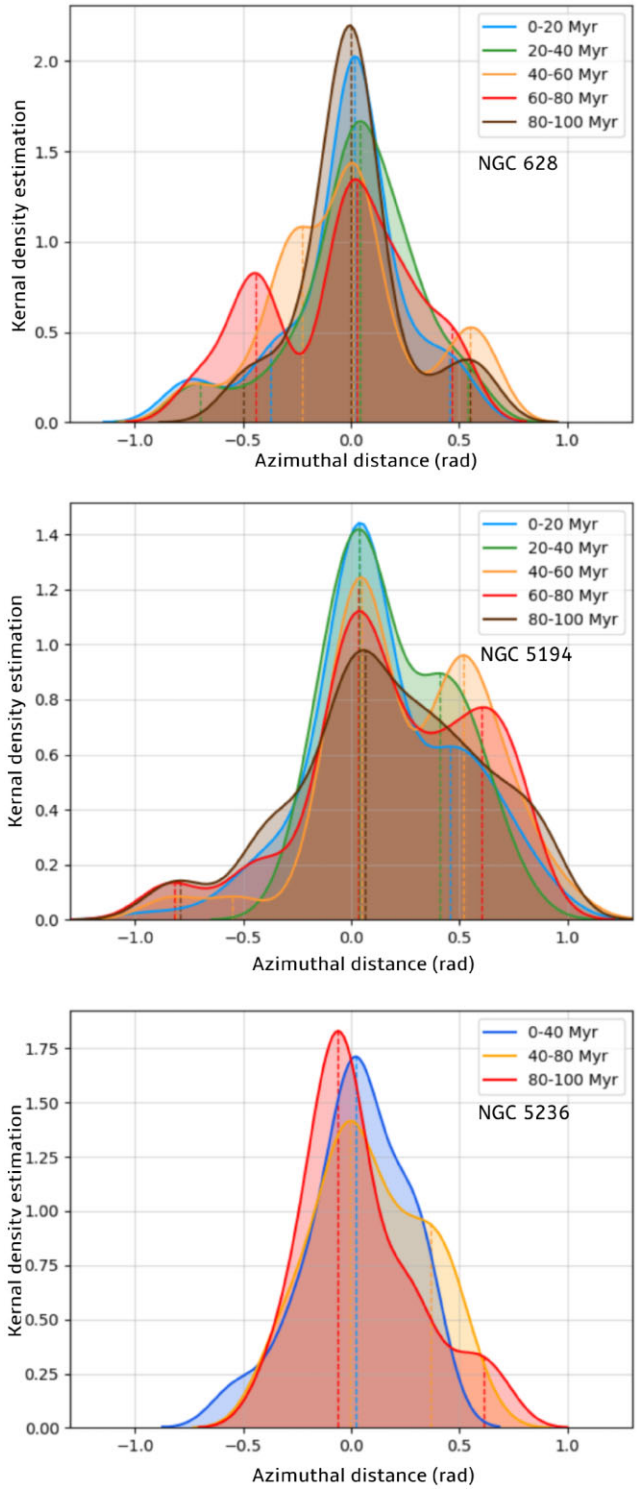

Figure 10. KDEs of the azimuthal cluster distance offset relative to the density wave. Dashed vertical lines coincide with the position of each peak. Top panel: NGC 628 analysed for the 0–100 Myr age range in 20 Myr age bins, plotted using an optimized bandwidth of 0.112 rad. Middle panel: NGC 5194 analysed for the 0–100 Myr age range in 20 Myr age bins, with a bandwidth of 0.131 rad. Bottom panel: The KDE of NGC 5236 in 0–40 Myr, 40–80 Myr, and 80–100 Myr age bins, plotted using an optimized bandwidth of 0.113 rad.

spatially distributed, spanning the inside and outside of the corotation radius. Our best estimate on the corotation radius for NGC 628 places R_{cr} at 4.5 ± 0.68 kpc, cf. 4.6 ± 1.2 kpc (Scarano & Lépine 2013).

Since disc particles move faster than the spiral arm inside R_{cr} and slower outside, it produces a $\pm\Delta\Theta$ in azimuthal measurements, giving rise to secondary peaks on either side of the spiral pattern zero location. In contrast to NGC 5194 and NGC 5236, in the case of NGC 628 (although most of the clusters are inside the R_{cr}) there is a significant number of clusters falling within the error margins and outside. This is more visible in the age bins 0–20 Myr and 80–100 Myr, which are consistent with the KDE plot (Fig. 10, top panel). Another possible reason may be simply due to the intrinsic nature of each galaxy. Grand designs may demonstrate clear evidence for age gradients in comparison to the others.

According to the Elmegreen & Elmegreen (1987) arm class classification, NGC 5194 is considered as an arm class 12 galaxy. Ideal examples for grand designs are considered to be of arm class 12, with two long symmetric arms dominating the optical disc. NGC 628 and NGC 5236, on the other hand, are both considered as arm class 9 galaxies, where two symmetric inner arms with multiple long and continuous outer arms are present. The asymmetries in the outer arms may cause the double peaks we see in certain age bins, but not in all. The gradual changes in the age gradients are also visible in cumulative KDEs. Fig. 11 (top panel), (middle panel) and (bottom panel) depict cumulative KDEs for NGC 628, NGC 5194, and NGC 5236, respectively.

5 SUMMARY AND CONCLUSIONS

The search for age gradients has been a quest for many scholars since the density wave theory was proposed by Lin & Shu (1964). Many have claimed to see compelling evidence for age gradients, while some have failed to see age gradients in galaxy structures. Since the existence of an age gradient is an implication of the density wave theory framework, the detection of an age gradient further bolsters the validity of the theory. Gonzalez & Graham (1996), Martínez-García et al. (2009), Davis et al. (2015), Yu & Ho (2018, 2019), Miller et al. (2019), and others have found evidence in favour of the density wave theory, while some, e.g. Schweizer (1976), Talbot et al. (1979), and Foyle et al. (2011) have found contradictory results.

The key points of our study are summarized as follows.

(i) Due to inconsistent results in age gradient studies found in the literature, we focused on SFH maps and stellar clusters rather than using the most common method, which involves the use of various physical tracers.

(ii) Observing SFH maps in different age bins showed an overall decrement (tightening) of pitch angles with increasing age, consistent with our previous studies (Pour-Imani et al. 2016; Miller et al. 2019; Abdeen et al. 2020). See Table 3 and Fig. 5.

(iii) The tightening of pitch angles with increasing age was further investigated by calculating the probability $P(\phi_1^\circ > \phi_2^\circ)$ that $\phi_1^\circ > \phi_2^\circ$ (see Table 4). By observing the last column of Table 4, it is visible that predominantly $\phi_1^\circ > \phi_2^\circ$. Considering the average of the entire sample for adjacent age bins, ϕ_2° is less than ϕ_1°, 69 per cent ± 25 per cent of the time.

(iv) A KDE plot for the entire sample is used to depict the pitch angle differences with respect to the 8 μm image in four different age bins. See Fig. 6.

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Table 5. Primary and secondary peak locations of the KDE plots for stellar clusters.

Galaxy	Age bin	Primary peak			Secondary peak	
		Peak location (rad)	max (KDE) (rad^{-1})	FWHM (rad)	Positive side (rad)	Negative side (rad)
	0–20 Myr	0.015	2.035	0.324	0.383	−0.372
	20–40 Myr	0.042	1.672	0.463	–	−0.699
NGC 628	40–60 Myr	0.011	1.441	0.540	0.553	−0.226
	60–80 Myr	0.024	1.349	0.870	0.465	−0.441
	80–100 Myr	−0.001	2.205	0.333	0.552	−0.497
	0–20 Myr	0.036	1.442	0.449	0.457	–
	20–40 Myr	0.038	1.420	0.764	0.408	–
NGC 5194	40–60 Myr	0.048	1.245	0.853	0.520	−0.548
	60–80 Myr	0.031	1.121	0.944	0.603	−0.467
	80–100 Myr	0.061	0.980	0.961	–	−0.793
	0–40 Myr	0.022	1.743	0.555	–	–
NGC 5236	40–80 Myr	−0.004	1.415	0.745	0.365	–
	80–100 Myr	−0.061	1.830	0.412	0.614	–

Notes. Columns: (1) Galaxy. (2) Age bin. (3) Primary peak location relative to the density wave (rad). (4) Maximum height of the KDE (rad^{-1}). (5) FWHM (rad). (6) Secondary peak location on the positive side of the density wave (rad). (7) Secondary peak location on the negative side of the density wave (rad).

(v) A clear age trend is visible when the brightest pixels selected from two young age bins of the NGC 5194 SFH map are superimposed on an 8 μm image. See Fig. 7.

(vi) The stellar cluster analysis revealed that the FWHM of the cluster distributions gradually increase with the cluster age, except for the 80–100 Myr age bin in NGC 628 and NGC 5236's 80–100 Myr age bin. See Table 5.

(vii) NGC 628 showed secondary peaks on both the positive and negative sides of the density wave. See Fig. 10 (top panel).

(viii) KDE plots revealed prominent central peaks and secondary peaks on the positive side of the density wave for NGC 5194 and NGC 5236. See Fig. 10 (middle panel) and (bottom panel).

Although we see a clear indication of a spatial age ordering in some galaxies, throughout the literature there are mixed opinions on this. One of the recent attempts in studying physical offsets between arm tracers via a meta-analysis of galaxies, Vallée (2020, table 1), summarizes the studies carried out over the previous 12 yr and tabulates instances where physical offsets were detected and cases where they failed to see notable offsets.[7] Considering NGC 5194 as an example, seven studies have shown positive physical age offsets, while six studies have failed to detect any. Vallée (2020) also indicated some of the possible reasons for the lack of detections, and they tend to be mainly due to the choice of tracers that are involved. We specifically selected the stellar cluster method in order to avoid this tracer biasing, by using stars with known age estimates.

Since we focused on the cluster ages, it is important to closely examine the related work of Shabani et al. (2018), as they too were looking at age gradients using stellar clusters and they did not detect a positive age offset for NGC 5194, while we did. One possibility for these inconsistent results may be due to the way we define the cluster locations relative to the density wave. We measure the cluster locations relative to logarithmic spiral arms placed over the brightest regions of the 8 μm images, to denote the shock location of the density wave. However, Shabani et al. (2018) have defined the spiral-arm ridge lines based on their meandering traces of the dark obscuring dust lanes on the *B*-band images. Although in general, the dark ridge lines in the *B*-band images do match with the 8 μm images, upon closely analysing these images using contour maps and pitch angle measurements, we are convinced that the choice of the spiral-arm location does result in the observed discrepancies between the two studies. Minor deviations from the spiral arms being not intrinsically logarithmic may also contribute to the observed results. The pros and cons of these choices may have to be closely examined in future studies.

Using cluster studies to check for the existence of age gradients, as has been done in Shabani et al. (2018) and Grasha et al. (2019), is a very promising method of probing density wave theory. A recent study by Bialopetravičius & Narbutis (2020) have also found evidence in favour of an age gradient by studying stellar clusters in NGC 5236, a galaxy which we have in common. They too were able to successfully detect an age gradient, thus further confirming our claim. Our experience with SFH maps suggests to us that age gradients are common in galaxies, and that studies using cluster positions confirm this in some cases. Each of these studies do also report cases where age gradients are not found, and it may be that some galaxies do not have them.

Bittner et al. (2017, 2020) show that grand designs appear to have significantly larger bulges than multi-armed galaxies. They provide a possible reasoning, explaining that a large central bulge might provide a high Toomre-*Q* criterion (Safronov 1960; Toomre 1964), capable of producing long-lasting, quasi-stationary density waves. Perhaps the existence of large central bulges may also play a role in the detection of age gradients.

Clearly, further study is needed, especially since some cluster studies are handicapped by relatively small numbers of cluster positions being available for clusters in the crucial 0–100 Myr category. Nevertheless, the existence of an age gradient in a considerable number of the galaxies we have studied is itself of great significance, even though we cannot yet conclusively answer the question of whether all or most galaxies exhibit signs of structure produced by long-lived density waves. We have clearly demonstrated that our method of age gradient analysis is effective in ascertaining the origin of volute structure in disc galaxies. As the sample size of galaxies studied is increased, further statistical significance will be acquired.

[7]More close to home, Griv et al. (2020) have conducted a study of the line-of-sight velocities of main-sequence stars from the extensive *Gaia* data base. Their findings show a predominant density wave structure in our local neighbourhood of the Galaxy.

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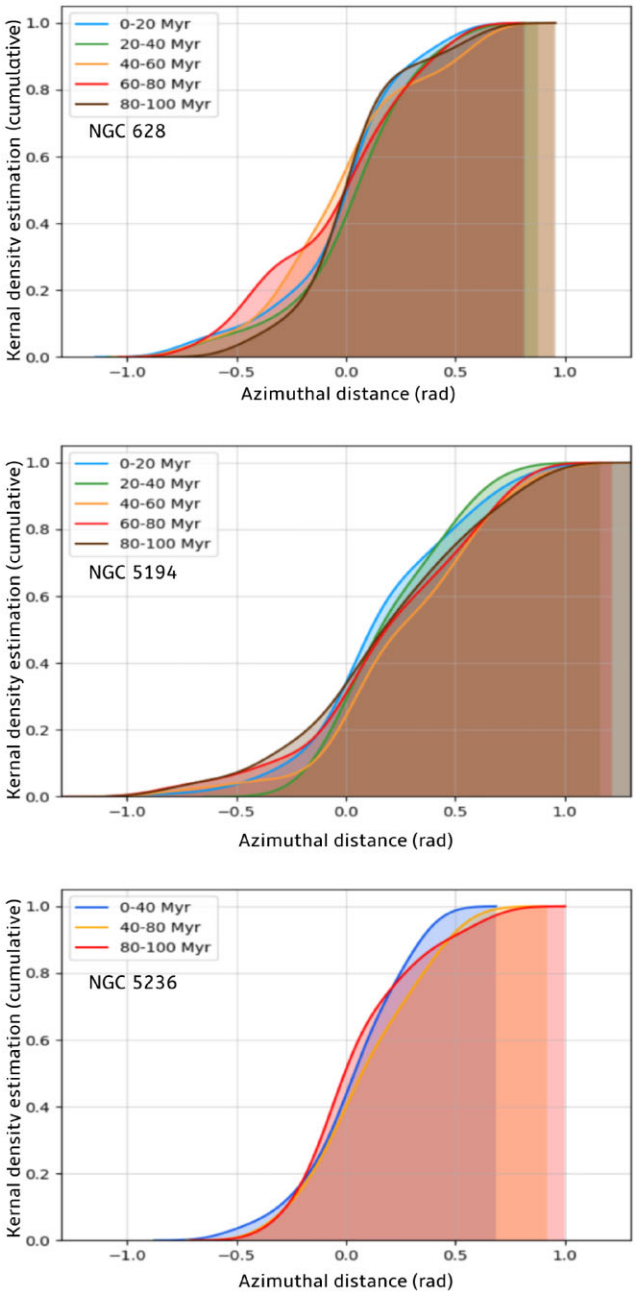

Figure 11. Cumulative KDEs of the azimuthal cluster distance offset relative to the density wave. Top panel: NGC 628 analysed from the 0–100 Myr age range in 20 Myr age bins, plotted using an optimized bandwidth of 0.112 rad. Middle panel: NGC 5194 analysed from the 0–100 Myr age range in 20 Myr age bins, with a bandwidth of 0.131 rad. Bottom panel: NGC 5236 in the 0–40 Myr, 40–80 Myr, and 80–100 Myr age bins, plotted using an optimized bandwidth of 0.113 rad.

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DATA AVAILABILITY

The data underlying this article will be shared upon reasonable request to the corresponding author.

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